

LIST

List is a structural element that is used to group or organize a collection of related items.

All the items in a particular list belong to the same category.

There are three types of list. They are:

1. Unordered List
2. Ordered List
3. Description List

1. Unordered List

Unordered List is a type of listing representation in which the order holds no meaning. Even if you jumble one list-value before another, it doesn't have any specific impact on the meaning.

It only has three different bullets: disc (black circle) , circle (hollow circle) and square. Additional like images could be added using CSS.

Syntax:

```
<ul>
    <li>List-Value</li>
    <li>List-Value</li>
    <li>List-Value</li>
</ul>
```

The below code shows a list of fruits. Order holds no meaning as they still belong to the same category 'fruit'. Even if Mango comes before Banana, they still belong to the same category.

```
<ul>
    <li>Banana</li>
    <li>Mango</li>
    <li>Apple</li>
    <li>Pineapple</li>
</ul>
```

Preview of Code:

- Banana
- Mango
- Apple
- Pineapple

By default the bulletins of an unordered list is a disc or black circle.

2. Ordered list

Ordered List is a type of listing representation in which the order holds some meaning (like a sequence or steps). Hence, the order itself shows the information.

It can have many bullet formats like decimal (1,2,3), upper-alpha (A, B), lower-alpha (a,b), upper-Roman (I, II), lower-Roman(i, ii) and so on.

Syntax:

```
<ol>
  <li>List-Value</li>
  <li>List-Value</li>
  <li>List-Value</li>
</ol>
```

The below code shows an ordered list. By default it uses 1,2,3, ...

```
<p>My daily morning routine includes:</p>
<ol>
  <li>Wake up</li>
  <li>Brush Teeth</li>
  <li>Exercise</li>
  <li>Drink Water</li>
  <li>Bath</li>
  <li>Go to college</li>
</ol>
```

My daily morning routine includes:

1. Wake up
2. Brush Teeth
3. Exercise
4. Drink Water
5. Bath
6. Go to college

Now if I swap the order of Brush Teeth with Wake up, the whole information changes. Hence, in this code the order itself is an information so it is shown in the following sequence (1,2,3, ..) and so on.

3. Description List

Description List is a type of listing representation which shows collection of description term and description detail in pairs.

It is used when you want to show additional information about the list values. Description List does not require any bullets but can be manually assigned by CSS and not HTML.

Syntax:

```
<dl> <!--Description List-->
  <dt>Description Term</dt>
  <dd>Description Details</dd>
```

```
<dt>Description Term</dt>
<dd>Description Details</dd>
</dl>
```

The below code shows an example of a description list.

```
<dl>
  <dt>ME</dt>
  <dd>It is a branch of engineering that offers study on mechanical field.</dd>

  <dt>EEE</dt>
  <dd>It is a branch of engineering that offers study on electrical field.</dd>

  <dt>CSE</dt>
  <dd>It is a branch of engineering that offers study on computer field.</dd>
</dl>
```

This list holds a more semantic meaning as it tells the web browser the relationship between list value (content) and its description.

Nested List

Nested List is a list that contains other lists within its list items.

Syntax:

```
<!-- Nested List -->
<ul>
  <li>Egg
    <ul>
      <li>Egg Curry</li>
      <li>Omlette</li>
      <li>Boiled Egg</li>
    </ul>
  </li>
  <li>Soybeans</li>
</ul>
```

This is the code preview.

- Egg
 - Egg Curry
 - Omlette
 - Boiled Egg
- Soybeans

You can insert a nested list by inserting another `<ul>` or `<ol>` tag before the end tag or `</li>` list item and after the content of `<li>` tag. This shows that this descendant list belongs to which list items. Don't place it after the end tag or `</li>` as it will mean there is an entirely new list without any relationship with the ancestor list values. Even if they show the same representation in browser it is semantically incorrect.

Incorrect Method:

```
<ul>
  <li>Egg</li>
  <ul>
    <li>Egg Curry</li>
    <li>Omlette</li>
  </ul>
  <li>Soybeans</li>
</ul>
```

You can implement multiple nested lists within nested lists and there is no limit.

Code:

```
<ul>
  <li>Egg
    <ul>
      <li>Egg Curry</li>
      <li>Omlette
        <ul>
          <li>1 egg</li>
          <li>2 eggs</li>
        </ul>
      </li>
    </ul>
  </li>
  <li>Soybeans</li>
</ul>
```

Preview:

- Egg
 - Egg Curry
 - Omlette
 - 1 egg
 - 2 eggs
 - Soybeans

See by default it will move from disc (black circle) to circle (white circle) to square.

Attributes of List

Only `<ol>` and `<ul>` tags have attributes. `<dl>` tag doesn't have even one.

1. start

- It sets the starting value.
- It is used only for `<ol>` tag and not `<ul>` tag.
- Code:

```
<ol start="5">
    <li>Wake up</li>
    <li>Brush Teeth</li>
    <li>Exercise</li>
    <li>Drink Water</li>
    <li>Bath</li>
    <li>Go to college</li>
</ol>
```

- Preview:

5. Wake up
6. Brush Teeth
7. Exercise
8. Drink Water
9. Bath
10. Go to college

2. type

- It changes the numbering style.
- It is used for both `<ol>` and `<ul>` tags.
- Code:

```
<ol type="A">
```

```
<li>Wake up</li>
<li>Brush Teeth</li>
<li>Exercise</li>
</ol>
```

- Preview:

- A. Wake up
- B. Brush Teeth
- C. Exercise

- Code:

```
<ul type="square">
  <li>Banana</li>
  <li>Mango</li>
  <li>Apple</li>
  <li>Pineapple</li>
</ul>
```

- Preview:

- Banana
- Mango
- Apple
- Pineapple

3. reversed

- It is used only for tag and not for tag.
- It counts backward.
- reversed is a boolean attribute and by default it is set to true if applied so no need to specify value like **reversed="true"** as it is the same as **reversed**.
- Code:

```
<ol type="A" reversed>
  <li>Wake up</li>
  <li>Brush Teeth</li>
  <li>Exercise</li>
```

```
</ol>
```

- Preview:

C. Wake up

B. Brush Teeth

A. Exercise